Enabling Semantic Traceability in Health Data: The Health-RI Semantic Interoperability Initiative

Pedro Paulo F. Barcelos[1], Niek van Ulzen[1], Reinier Groeneveld[1], Ana Konrad[1], Qasim Khalid[2], Shuxin Zhang[3], Annemarie Trompert[1], and Janet Vos[1]

health RI
enabling data driven health

BACKGROUND AND MOTIVATION

Fragmented landscape

- Health data are distributed across many standards, data models, and local schemas
- The same domain concepts are represented differently across care, research, and infrastructure settings

Semantic gap

- Data may be exchanged syntactically while meaning is not preserved across systems
- Similar labels, codes, or formal structures do not guarantee the same conceptualization
- This may create false agreement: systems may appear aligned while remaining semantically misaligned

Limits of current practice

- Interoperability still depends heavily on manual, case-by-case pairwise mappings
- These mappings are difficult to scale, maintain, and revise as standards and local schemas evolve
- Meaning-level assumptions often remain implicit, incomplete, and hard to compare across artifacts

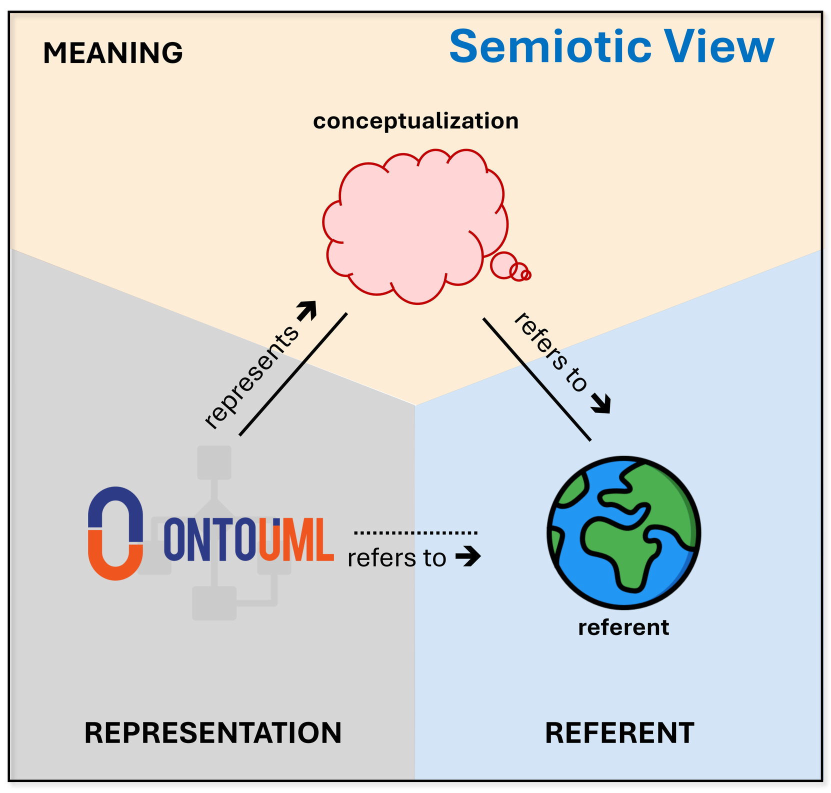
What is needed

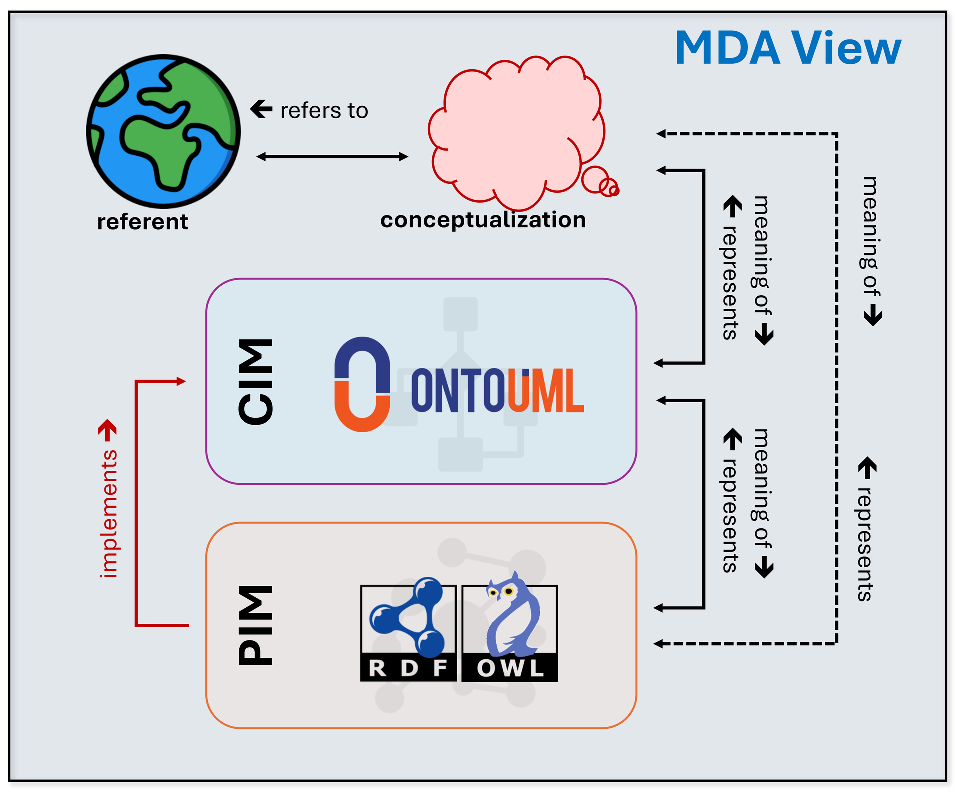
- A common semantic reference model is needed to make intended meaning explicit and traceable across artifacts
- Standards and local schemas should connect to this shared reference through reusable mappings, rather than being repeatedly reconciled pairwise

Why this matters

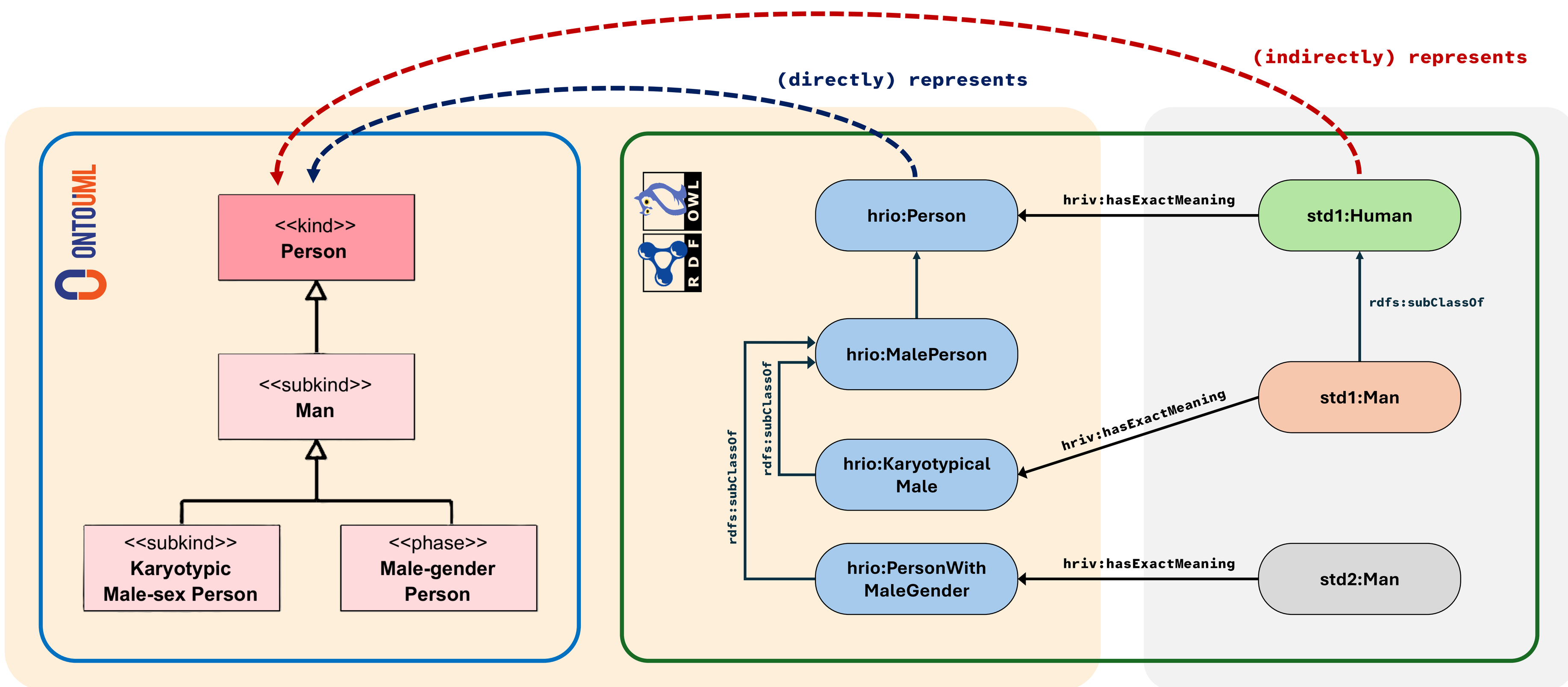
- Clearer semantic alignment can help reduce misinterpretation, mitigate false agreement, and support more trustworthy integration
- It can improve comparability, reuse, and FAIR-aligned interoperability across institutions and use cases

ENABLING SEMANTIC TRACEABILITY





Semantic traceability in the Health-RI Semantic Interoperability Initiative's architecture. The semiotic view (left) shows OntoUML as a representation of a shared conceptualization of the referent domain. The MDA view (right) shows HRIO OntoUML as the CIM and the HRIO gUFO/OWL ontology as the PIM, with the PIM implementing the CIM.



Example of semantic traceability: a fragment of HRIO OntoUML for Person's sex and gender is implemented in HRIO gUFO/OWL, and external ontologies' OWL classes are aligned via HRIV mappings, distinguishing direct and indirect representation relations. The semantics of the external classes are traceable to their corresponding meanings modeled in OntoUML.

ARTIFACTS

REFERENCES

https://w3id.org/health-ri/semantic-interoperability

Barcelos, P. P. F., van Ulzen, N., Groeneveld, R., Konrad, A., Khalid, Q., Zhang, S., Trompert, A., & Vos, J. (2026, March 24–25). *Enabling semantic traceability in health data: The Health-RI Semantic Interoperability Initiative.* SWAT4HCLS 2026: Semantic Web Applications and Tools for Health Care and Life Sciences, Amsterdam, the Netherlands.

[1] Health-RI, Jaarbeursplein 6, 3521 AL, Utrecht, The Netherlands
[2] Leiden University Medical Center (LUMC), Albinusdreef 2, 2333 ZA Leiden, The Netherlands
[3] Amsterdam University Medical Center, Meibergdreef 9, 1105 AZ, Amsterdam-Zuidoost, The Netherlands